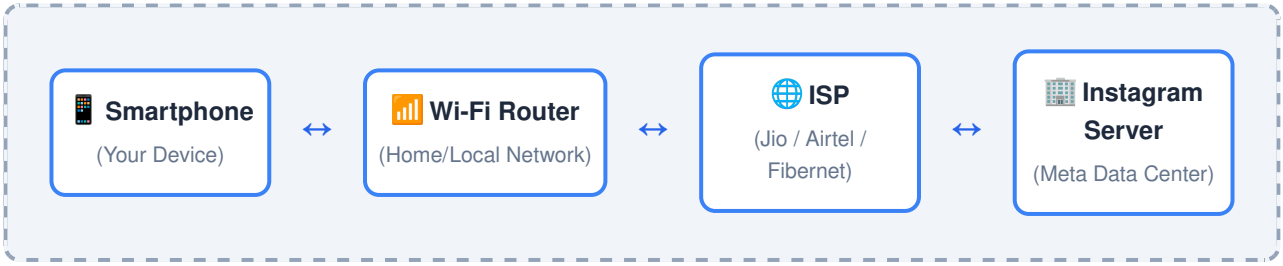
# Computer Networking Assignment

Solutions to Questions 1–4 | Detailed & Styled Reference

## Q1. Network Connection Diagram (Smartphone to Instagram Server)

Below is a diagram illustrating how data flows from your smartphone to Instagram's remote servers and back across various networking components:




**Detailed Connection Breakdown & Flow:**

1. **Smartphone (Device):** Generates the data request (e.g., loading feed or images) via the Instagram app using wireless signals (Wi-Fi or Mobile Data).
2. **Wi-Fi Router (Local Access Point):** Receives wireless radio signals from your phone, converts them into digital network packets, and routes them via Ethernet/Fiber connection to the modem/ISP line.
3. **ISP (Internet Service Provider):** Companies like Jio, Airtel, or BSNL route your data packets through their large-scale optic fiber cables across the internet backbone.
4. **Instagram Server (Destination):** Meta's cloud server receives your request, processes it, and sends the requested photos/videos back through the same path to your phone screen.

## Q2. Daily Networked Devices and Network Types

Here are 5 common devices used daily, along with the specific network types they connect to:

| Device | Network Type | Description / Connection Details |
|---|---|---|
| **1. Smartwatch** | **PAN** (Personal Area Network) | Connects directly to a smartphone over short distances using **Bluetooth**. |
| **2. Laptop** | **WLAN / LAN** (Local Area Network) | Connects wirelessly via Wi-Fi or wired Ethernet to the home/office router. |

| Device | Network Type | Description / Connection Details |
| --- | --- | --- |
| **3. Smart TV** | LAN (Local Area Network) | Connects to the local home Wi-Fi network to stream videos from local or web media. |
| **4. Wi-Fi Router** | WAN (Wide Area Network) | Connects the local network (home devices) to the global Internet provided by the ISP. |
| **5. Smartphone** | WAN / Cellular | Connects to wide geographical cellular towers (4G/5G) or Wi-Fi networks. |

## Q3. Difference Between Internet and Intranet

The key differences between the public Internet and a private Intranet are highlighted below:

| Feature | The Internet | An Intranet |
| --- | --- | --- |
| **Definition** | A global public network of interconnected computers accessible by anyone worldwide. | A private, restricted network accessible only to authorized members of an organization. |
| **Access** | Public (Open to everyone). | Private (Requires specific logins/ network connection). |
| **Security** | Lower security, exposed to public traffic. | High security behind firewalls and authentication gateways. |

> **Real-World Example (College / Campus Wi-Fi):**
> When you connect to your **College Wi-Fi**, accessing the college internal portal (e.g., student attendance system, internal grade portal, or library database) works on the **Intranet** (only accessible inside campus). However, when you open `google.com` or `youtube.com` using the same Wi-Fi, you are accessing the global **Internet**.

## Q4. Data Journey During an In-App Search (e.g., Zomato / Amazon)

**Selected App: Zomato (Food Delivery App)**

Here is how data travels in 4–5 concise steps when you hit 'Search' inside the app:

1. **Request Trigger:** When you type *"Pizza"* and hit 'Search', your smartphone packages this search query into an HTTP/HTTPS data packet.

2. **Local Transmission:** The smartphone transmits this request packet wirelessly via Wi-Fi or cellular towers (4G/5G) to your ISP (Internet Service Provider).
3. **Internet Routing:** The ISP routes the packet across the global Internet backbone directly to Zomato’s cloud database servers.
4. **Server Processing:** Zomato's server processes the request, queries its database for nearby pizza restaurants, and compiles the search results into a JSON response packet.
5. **Return Journey:** The server sends this response back through the ISP to your Wi-Fi router/cellular tower, displaying the list of pizza places instantly on your smartphone screen.